

FINAL PROJECT REPORT

RecipeGrid

An Interactive Website

Theme: Digital Recipe Book

Recipe search, Dynamic content, User submitted recipes

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1. Project Overview

1.1 Project Purpose

RecipeGrid is a database-driven recipe website that makes it easy to find, organize and share recipes. Users can browse and search recipes in one place, filter them by category to decide what to cook faster, and upload their own recipes for others to enjoy.

1.2 Target Audience

- University Students – students living away from home who need fast, easy-to-follow, and budget-friendly recipe guides suited to limited ingredients and time constraints.
- Home Cooks & Food Enthusiasts – individuals who want a structured environment to document culinary discoveries, maintain a personalized cooking ledger, and share content with a wider community.
- Health-Conscious Individuals – users looking for structured dietary filters such as vegetarian alternatives or nutrient-dense quick meals.

1.3 Key Features

Feature	Proposal Description
Dynamic Recipe Search & Filtering	Search and filter the recipe directory by title, or predefined cooking categories.
Secure User Authentication System	User signup, secure verification, and session termination.
User-Submitted Recipe Dashboard	Authenticated users can submit new recipes through a structured insertion form.
Interactive Contact & Feedback System	Users can submit inquiries, technical feedback, or feature requests.

1.4 Technology Stack

Layer	Technology
Frontend	HTML5, CSS3, Bootstrap 5
Client-side Logic	Vanilla JavaScript
Backend	PHP 8
Database	MySQL (via XAMPP)
Version Control	Git & GitHub

1.5 Project Repository

GitHub Repository: [Click here to access the project on GitHub](#)

2. System Design

2.1 Database Schema

users(id, full_name, email, password, created_at)

Primary Key: id

UNIQUE: email

recipes(id, user_id, title, category, prep_time, description, image,
ingredients, instructions, created_at)

Primary Key: id

Foreign Key: user_id → users(id)

ON DELETE SET NULL, ON UPDATE CASCADE

Recipe Database ER Diagram

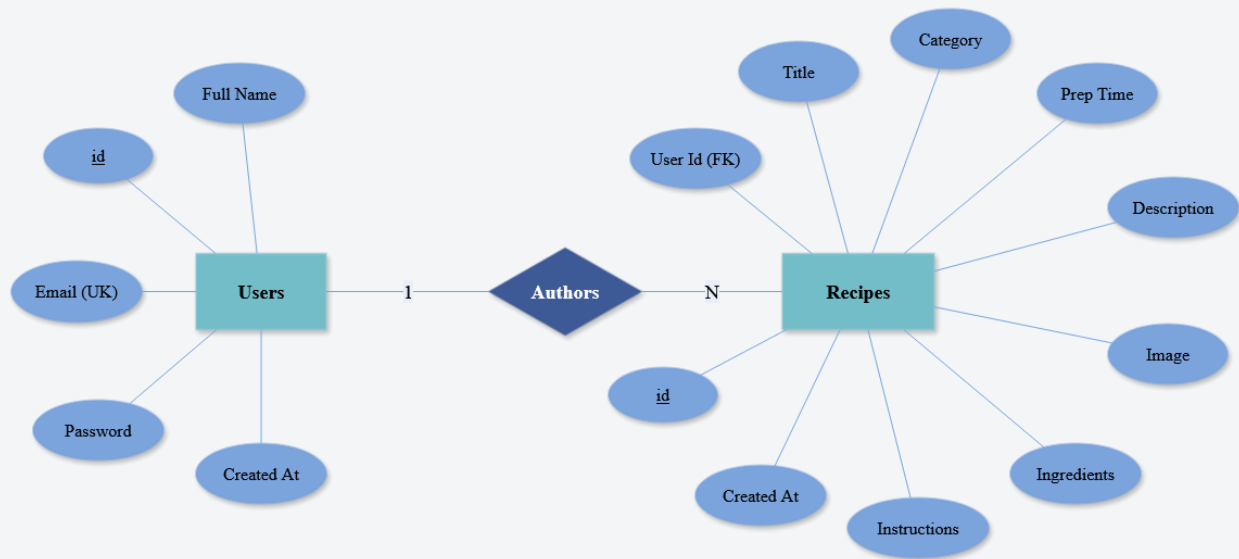


Figure 1-ER Diagram

2.2 Database Table Summary

Table	Purpose	Primary Key	Foreign Keys	Important Fields
Users	Stores registered user accounts and their login credentials.	<i>id</i>	None	<i>full_name</i> , <i>email</i> (unique), <i>password</i> (stores the hashed password), <i>created_at</i>
Recipes	Stores recipes shared on the platform, including their details and preparation steps.	<i>id</i>	<i>user_id</i> references users(id)	<i>title</i> , <i>category</i> , <i>prep_time</i> , <i>description</i> , <i>image</i> , <i>ingredients</i> , <i>instructions</i> ,

				<i>created_at</i>
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2.3 Folder Structure

```

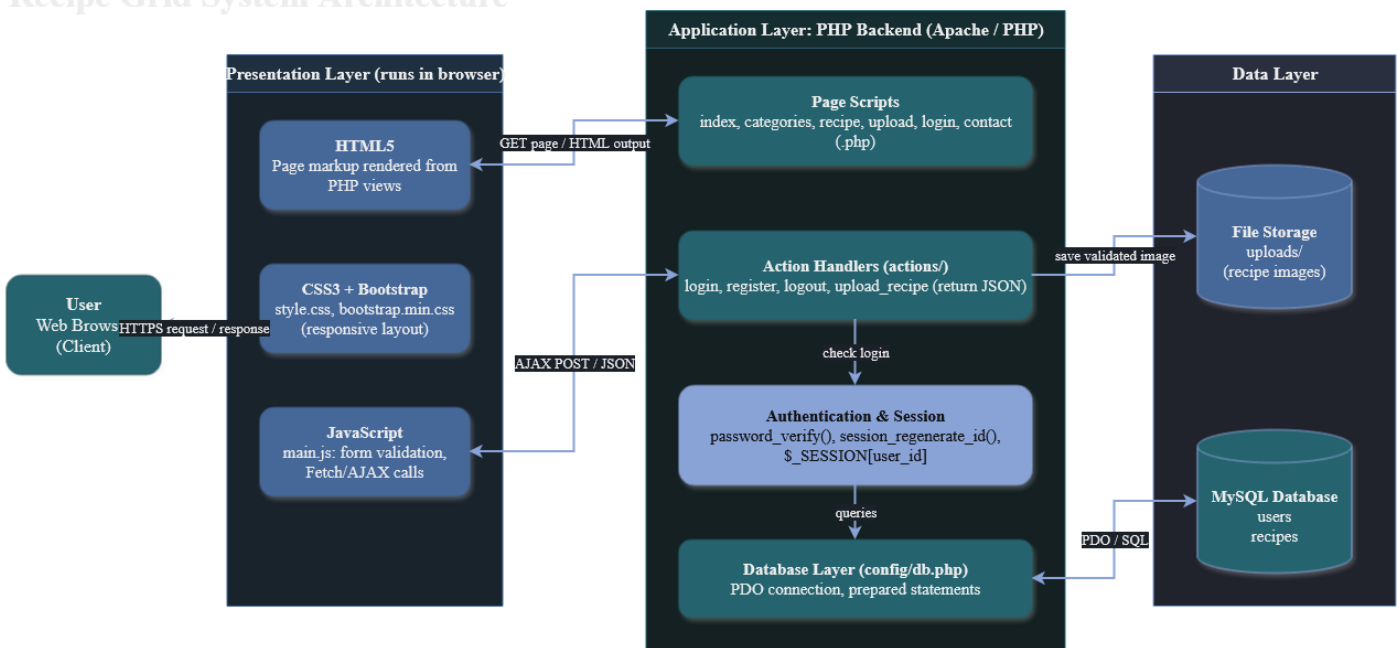
/
└─ ict1209_group36_recipe-grid
    └─ actions
        ├── login.php
        ├── logout.php
        ├── register.php
        └─ upload_recipe.php
    └─ config
        └─ db.php
    └─ css
        ├── bootstrap.min.css
        └─ style.css
    └─ images
    └─ js
        └─ main.js
    └─ uploads
    ├── index.php
    ├── categories.php
    ├── recipe.php
    ├── login.php
    ├── upload.php
    ├── contact.php
    ├── recipe_grid.sql
    └─ README.md

```

2.4 System Architecture

The Recipe Grid system uses a layered architecture. The user's browser (HTML, CSS, Bootstrap and JavaScript) sends requests to a PHP backend, which reads and writes data in a MySQL database through PDO. Uploaded images are stored in the uploads folder. Login is handled with PHP sessions: the password is checked against its stored hash, and the user's ID is saved in the session so that protected actions, such as uploading a recipe, are available only to logged-in users.

Recipe Grid System Architecture



Authentication flow: (1) The login form is submitted by JavaScript as an AJAX POST to actions/login.php. (2) PHP looks up the user with a prepared statement and checks the password with password_verify(). (3) On success the session ID is regenerated and user_id is stored in \$_SESSION. (4) Protected actions such as upload_recipe.php check \$_SESSION[user_id] before saving the image to uploads/ and the recipe to MySQL.

Figure 2-System Architecture

3. Implementation

3.1 Dynamic Recipe Search & Filtering

The system provides a recipe directory that users can scan and filter by recipe title and predefined cooking categories.

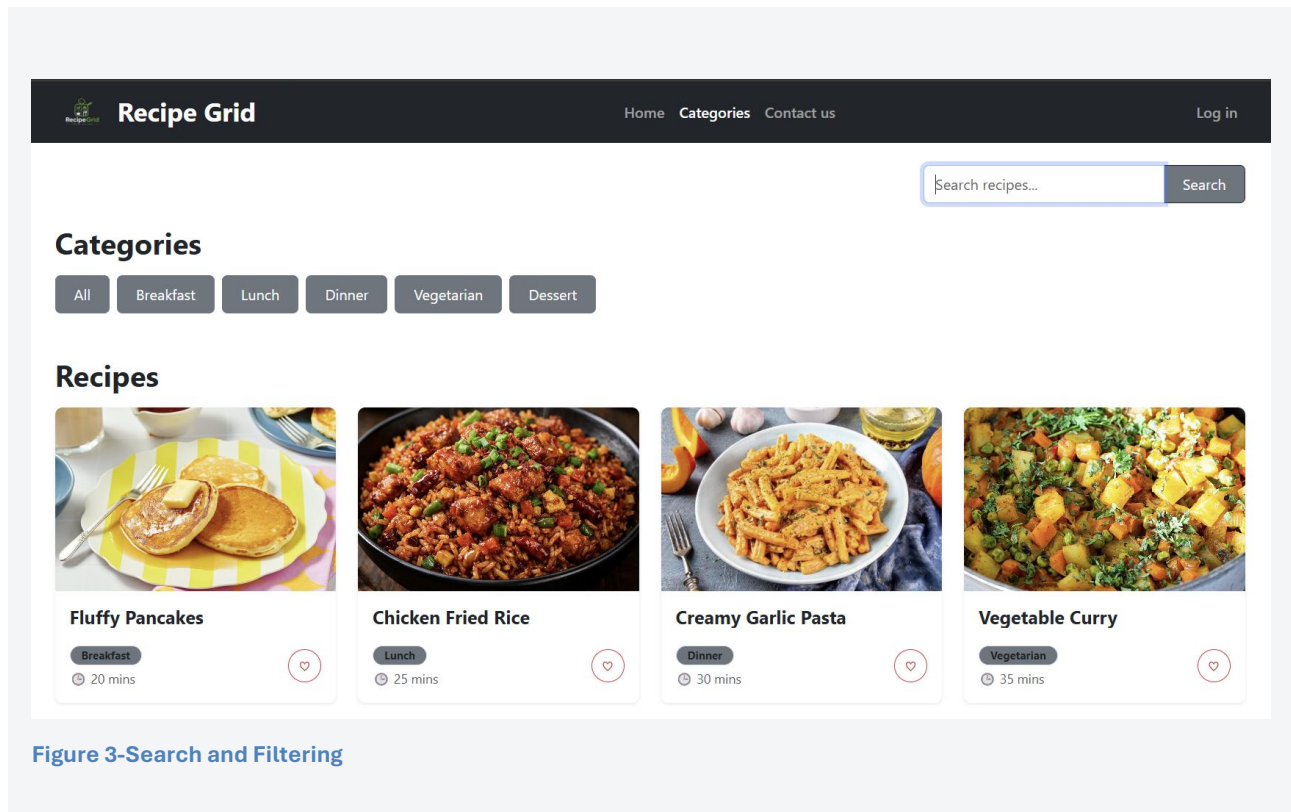


Figure 3-Search and Filtering

Search and filtering are built into the Categories page and update results instantly without reloading the page. Users can search by recipe name or filter by category. The search is partial-match, so typing even one correct word is enough to find a recipe without entering its full name.

3.2 Secure User Authentication System

The authentication component includes user signup, verification, and session termination.

The figure displays two side-by-side authentication forms within a light gray container. The left form is titled 'Create Account' and features a sub-header 'Join Recipe Grid to save and share recipes.' It includes input fields for 'Full Name' (with a placeholder 'Name'), 'Email address' (with a placeholder 'name@example.com'), and 'Password' (with a placeholder '*****'). A 'Create Account' button is at the bottom. The right form is titled 'Welcome Back' and features a sub-header 'Enter your details to log into your account.' It includes input fields for 'Email address' (with a placeholder 'name@example.com') and 'Password' (with a placeholder '*****'). A 'Log In' button is at the bottom. Both forms have 'Log In' and 'Sign Up' buttons at the top, with the 'Sign Up' button highlighted in blue on the left and the 'Log In' button highlighted in blue on the right. A 'Forgot password?' link is located below the password field on the right form.

Figure 4-Sign Up and Log In (empty)

Final Authentication Flow

A new user registers with a full name, email and password. The system stores the password only as a secure hash. On login, the submitted password is checked against the stored hash. If it matches, a new session is created and the user's ID and name are saved in it. Protected pages and actions, such as recipe upload, check this session before they run. On logout, the session data is cleared, the session cookie is removed and the user is returned to the home page.

Validation Behavior

Forms are validated twice, in the browser and again on the server. The browser checks that required fields are filled and shows Bootstrap error messages. The server then checks the following:

- The full name is required and limited to 100 characters.
- The email must be in a valid format.

- The password must be at least 6 characters.
- A duplicate email is rejected with the message "Email already registered."
- A wrong email or password shows one generic error, "Invalid email or password," so the system does not reveal which one was incorrect.

Log In
Sign Up

Invalid email or password.

Create Account

Join Recipe Grid to save and share recipes.

Full Name

 ✓

Email address

 ⓘ

Please enter a valid email address.

Password

 ⓘ

Password must be at least 6 characters.

Create Account

Log In
Sign Up

Invalid email or password.

Welcome Back

Enter your details to log into your account.

Email address

Password

[Forgot password?](#)

Log In

Figure 5-Sign Up and Log In (Invalid)

3.3 User-Submitted Recipe Dashboard

Authenticated accounts can access a structured form for inserting new recipes into the database.

Share Your Recipe with The World

Fill in the details below to publish your delicious creation onto Recipe Grid.

Recipe Title

e.g., Mix fried rice

Category

Select Category

Cooking Time (minutes)

e.g., 25

Short Description

A brief summary of your dish...

Recipe Cover Image

Choose File

No file chosen

Accepted formats: JPG, PNG, WEBP (Max size: 5MB).

Ingredients

List each ingredient on a new line
e.g.,
200g Pasta
2 cloves Garlic

Step-by-Step Instructions

Detail the preparation steps numbered or line-by-line...

Publish Recipe

Figure 6-Recipe Upload

The Recipe Upload page lets logged-in users publish their own recipes through a single form. Users enter a title, select a category, cooking time, short description, cover image, ingredients and step-by-step instructions. The cover image must be a JPG, PNG or WEBP file of up to 5MB. The form is validated in the browser and again on the server before the image is saved to the uploads folder and the recipe is stored in the database.

3.4 Interactive Contact & Feedback System

The contact and feedback interface allows clients/users to submit inquiries, technical feedback, or feature requests.

Contact Us

We'd love to hear from you. Whether you have a question, suggestion, or just want to say hello, feel free to reach out to us.

About Us

At Recipe Grid, we believe that great food does far more than just nourish—it brings people together and turns ordinary moments into celebrations. Cooking should be an inviting, joyful experience from the moment you look for inspiration to the very last bite. Our mission is to empower your culinary journey with easy-to-follow, reliable recipes crafted for every skill level, dietary need, and occasion.

Whether you need quick weeknight dinners, wholesome everyday meals, or decadent desserts, Recipe Grid is designed to help you find the perfect dish in seconds. Our visual layout allows you to effortlessly browse vibrant photos and organized categories, eliminating guesswork with clear step-by-step instructions. Recipe Grid is here to make your time in the kitchen simple, fun, and delicious every day.

Get in Touch

+94 70 333 336

Email: recipegrid77@gmail.com

Send a Message

Name

Phone Number

Email

Subject

Message

0 / 300 characters

Send Message

Figure 7-Contact

The Contact Us page tells visitors about Recipe Grid and gives them a way to get in touch. The left side has a short "About Us" section with the site's mission, plus the phone number and email address. The right side has a message form with name, phone number, email, subject and message fields. The message is limited to 300 characters, with a live character counter. The form is validated in the browser, and a confirmation message appears after a valid submission. *The contact form is client side only.

4. Challenges and Solutions

Challenge	Cause	Solution Implemented	Outcome
Database connection and import errors	The db.php settings (database name, username, password) did not match the local MySQL setup. Importing recipe_grid.sql also failed at first because the recipes table was created before the users table it depends on.	Corrected the connection details in config/db.php and reordered the SQL script so users is created before recipes. Added a try/catch around the PDO connection so errors are logged instead of breaking the page.	The database imports cleanly and every page loads data without connection errors.
Login did not work even with the correct details	Test users were first added to the database with plain-text passwords, but the login code compares the password against a stored hash. Sessions also failed at first because session_start() was not called at the top of some pages.	Registered users through the signup form so passwords are saved with password_hash(), and checked them with password_verify(). Added session_start() at the top of every page and handler.	Registered users can log in and stay logged in across pages, and logout ends the session properly.
Recipes could not be uploaded	Images larger than PHP's default upload limit were silently rejected, and the uploads folder was missing or not writable. Only some form fields were being checked, so incomplete data caused database errors.	Added server-side checks for image size (5MB), file type (JPG, PNG, WEBP) and all required fields. Created the uploads folder automatically with write permission, gave each image a unique name, and returned clear error messages.	Recipes with images now upload successfully. Invalid or incomplete submissions are rejected with a clear message instead of failing silently.

4.1 Testing and Fixes

Test / Issue	Expected Result	Actual Result	Fix / Status
Register with an email that already exists	System rejects the signup and shows "Email already registered."	A second account with the same email was created in an early version.	Added an email check before saving, plus a unique key on email in the database. Fixed and retested.
Log in with a registered account	User is logged in and taken to the home page.	Login failed with the correct password because test users had plain-text passwords in the database.	Registered users through the signup form so passwords are hashed, and used password_verify() at login. Fixed.
Upload a recipe with a large or wrong-format image	Upload is rejected with a clear error message.	The upload failed silently and no message was shown.	Added file size (5MB) and file type checks, and returned JSON error messages shown in an alert. Fixed.
Search recipes by one word on the Categories page	Matching recipes appear instantly without reloading the page.	Results appeared correctly for partial words and category filters.	No fix needed. Passed.
Open the upload page or log out while logged out	Upload is blocked for logged-out users, and logout ends the session.	Logged-out users could reach the upload form, and the session stayed active after logout.	Added a session check on upload and cleared the session and cookie on logout. Fixed.

5. Individual Contribution

Member	Role / Responsibility	Specific Tasks Completed	Evidence / Files / Screens
H.M.S. Premakumara ITT/2024/084 Index 2769	Frontend and backend developer, database integrator, tester	Frontend coding, backend coding and fixes, database integration, documentation, testing and debugging	actions/ (login, register, logout, upload_recipe), config/db.php, recipe_grid.sql, upload.php, testing and fixes tables, report documentation
M.A.V.R. PERERA ITT/2024/081 Index 2766	Planner, UI designer, client-side developer, tester	Planning, frontend design and UI, backend design, client-side coding in JavaScript, testing of the interface and forms	Page layouts and design (index.php, categories.php, recipe.php), css/style.css, js/main.js (form validation, AJAX, search and filtering), planning notes, test results for the UI and forms

5.1 Contribution Summary

Work Area	H.M.S. Premakumara	M.A.V.R. Perera
Planning / analysis	Helped with requirements and the project scope	Led the planning and requirement analysis
UI / frontend	Frontend coding and page integration	Frontend design and UI layout
JavaScript / client-side	Connected forms to the PHP handlers	Wrote the client-side JavaScript (validation, AJAX, search)
PHP / backend	Backend coding, login, registration and upload handlers,	Backend design and structure

	and fixes	
Database / SQL	Database integration and SQL script	Helped with the table design
Testing / debugging	Tested the backend, database and upload features, and fixed bugs	Tested the interface, forms and client-side features
Documentation / presentation	Created the project proposal and report.	Provided content and design material for the report and presentation

6. Setup Instructions

6.1 Requirements

- XAMPP with Apache, MySQL/MariaDB and PHP 8
- A web browser (Chrome, Edge or Firefox)
- The final RecipeGrid source-code folder: ict1209_group36_recipe-grid
- The database export recipe_grid.sql, located in the project root folder

6.2 Installation and Database Import

1. Install XAMPP and open the XAMPP Control Panel.
2. Start the **Apache** and **MySQL** services. Both should show a green "Running" status.
3. Copy the project folder ict1209_group36_recipe-grid into the XAMPP htdocs directory:
Windows: C:\xampp\htdocs\
macOS: /Applications/XAMPP/htdocs/
4. Open phpMyAdmin at <http://localhost/phpmyadmin>.
5. Click the **Import** tab, choose recipe_grid.sql from the project root, and click **Import**. This creates the recipe_grid database with the users and recipes tables and adds the sample recipes.

6.3 Running the Application

6. Open a web browser.
7. Go to http://localhost/ict1209_group36_recipe-grid/. The folder name in the URL must match the folder name in htdocs.

8. Confirm that the home page loads correctly.

9. Test the main features:

Search and filtering: open the Categories page, type part of a recipe name, and choose a category. The results should update without a page reload.

Authentication: click Log in, switch to Sign Up, and create an account. Try an invalid email and a password shorter than 6 characters to see the validation messages. Then log in with the new account.

Recipe submission: while logged in, open the Upload page, fill in all the fields, choose a JPG, PNG or WEBP image under 5MB, and click Publish Recipe. The new recipe should appear on the site.

Contact form: open the Contact page, fill in the form, and click Send Message. The character counter should update as you type, and a confirmation message should appear after a valid submission.

Logout: log out and confirm that the Upload page can no longer be used.

7. References

Project Proposal. Group 36 – RecipeGrid project proposal.

GitHub Repository. RecipeGrid source repository.

Repository: https://github.com/itt2024084-Suneth/ict1209_group36_recipe-grid.git